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Reference Data Setup

In this page you can find a set of suggestions for Reference Data monitoring using Grafana.

Suggestions



There are VM resource related metrics (CPU, Memory, IO, etc) that can also be added to Grafana. These were left out on purpose Google already has out of the box monitoring support for these resources.

These suggestions can be either new dashboards, changes to existing dashboards or completely new dashboard pages. Please find the metric description in [Reference Data Top Metrics](#).

Cluster page

Currently there is only one page where we can filter metrics by each of the instances. Although useful, having a cluster view monitoring page gives a view of how the overall cluster health. Given this, our suggestion is to add a new page with the following metrics:

Row Title	Title	Description	Query Details
Healthchecks - RDS	Reference Data Service	Displays reference data server healthcheck. The outcome is a bool, meaning that it assumes the value one when everything is okay with the service itself, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'reference-data-server' , \$timeFilter Group By: time(\$__interval), "host"
Healthchecks - RDS	Deadlocks	Displays deadlocks healthcheck. This healthcheck is carried out by reference data service in order to validate if some of its threads entered in deadlock. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'deadlocks' , \$timeFilter Group By: time(\$__interval), "host"
Healthchecks - RDS	Entity Repository	Displays entity repository healthcheck. This healthcheck is carried out by reference data service in order to validate if it is able to reach entity repository, which is an abstraction for a non-sql db. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'entity-repository' , \$timeFilter Group By: time(\$__interval), "host"
HTTP Requests	HTTP Throughput	Measures reference-data HTTP throughput.	Query A Field: "count"

Row Title	Title	Description	Query Details
			From: "io.dropwizard.jetty.MutableServletContextHandler.requests" Where: \$timeFilter Group By: time(\$interval), "host"
HTTP Requests	HTTP Throughput per Method	Measures reference-data HTTP throughput per method.	Query A Field: "count" From: "io.dropwizard.jetty.MutableServletContextHandler.get-requests" Where: \$timeFilter Group By: time(\$interval), "host"
HTTP Requests	2XX Requests Rate	Measures reference-data HTTP throughput with return code falls into 2xx.	Query A Field: "count" From: "io.dropwizard.jetty.MutableServletContextHandler.2xx-responses" Where: \$timeFilter Group By: time(\$interval), "host"
HTTP Requests	4XX Errors	Measures the number of 4xx errors returned by reference-data.	Query A Field: "count" From: "io.dropwizard.jetty.MutableServletContextHandler.4xx-responses" Where: \$timeFilter Group By: time(\$interval), "host"
HTTP Requests	5XX Errors	Measures the number of 5xx errors returned by reference-data.	Query A Field: "count" From: "io.dropwizard.jetty.MutableServletContextHandler.5xx-responses" Where: \$timeFilter Group By: time(\$interval), "host"
Global Database Requests	Global Database Throughput	Measures the overall database throughput, considering all operations executed by reference-data. It also shows a break down per operation.	Query A Field: "count" From: "reference-data-service.entity-repository" Where: \$timeFilter Group By: time(\$interval), "host", "operation"
Global Database Requests	Global Database Latency at Percentile 99	Measures global database latency at percentile 99 considering all operations executed by reference-data.	Query A Field: "p99" From: "reference-data-service.entity-repository" Where: \$timeFilter Group By: time(\$__interval), "host", "operation"
Global Database Requests	Global Database Errors	Measures the total number of errors registered while performing some operations onto the database.	Query A Field: "count" From: "reference-data-service.entity-repository.error" Where: \$timeFilter Group By: time(\$__interval), "host"
Global Database Requests	Global Database Errors - Entities Not Found	Measures the total number of errors caused by entities not found, registered while performing some operations onto the database.	Query A Field: "count" From: "reference-data-service.entity-repository.entitynotfound" Where: \$timeFilter Group By: time(\$__interval), "host"
Database Operations	Database Throughput Per Entity	Measures reference-data throughput per entity.	Query A Field: "count" From: "reference-data-service.entity-repository" Where: \$timeFilter Group By: time(\$interval), "host", "entityType", "operation"
	Global Database Latency at	Measures database latency at percentile 99 per entity,	Query A Field: "p99"

Row Title	Title	Description	Query Details
Database Operations	Percentile 99 per Entity	considering all operations executed by reference-data.	From: "reference-data-service.entity-repository" Where: \$timeFilter Group By: time(\$__interval), "host", "entityType", "operation"
JVM	JVM Heap Usage	Measures the jvm heap usage per component instance.	Query A Field: "value" From: "letmeknow.jvm.memory.heap.usage" Where: \$timeFilter Group By: time(\$__interval), "host"
JVM	JVM Heap Usage After GC	Measures the jvm heap usage after gc per instance.	Query A Field: "value" From: "letmeknow.jvm.memory.heap.after-gc.usage" Where: \$timeFilter Group By: time(\$__interval), "host"
JVM	JVM Garbage Collector Time	Measures maximum JVM GC duration. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	Query A Field: "value" From: "letmeknow.jvm.gc.G1-Old-Generation.time" Where: \$timeFilter Group By: "host"
JVM	JVM Active Threads Count	Measures the current number of active threads in the JVM per instance.	Query A Field: "value" From: "letmeknow.jvm.threads.count" Where: \$timeFilter Group By: time(\$__interval), "host"
JVM	JVM File Descriptors Ratio	Measures the jvm file descriptors usage ratio per instance.	Query A Field: "value" From: "letmeknow.jvm.filedescriptor" Where: \$timeFilter Group By: time(\$__interval), "host"

The **Row Title** column refers to a Grafana feature to group dashboards by type. This is only a suggestion and thus not mandatory.

Node page🔗🔗

These suggestions relate to the existing Reference Data dashboard page. You may find some new dashboards or changes to existing ones.

Our suggestions are to add the dashboards in the following table with a filter by each of the available nodes and applications:

Row Title	Title	Description	Query Details
Healthchecks - RDS	Reference Data Service	Displays reference data server healthcheck. The outcome is a bool, meaning that it assumes the value one when everything is okay with the service itself, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'reference-data-server' , "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
Healthchecks - RDS	Deadlocks	Displays deadlocks healthcheck. This healthcheck is carried out by reference data service in order to validate if some of its threads entered in deadlock. The outcome is a bool, meaning that it was	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'deadlocks' , "host" = '\$host' , \$timeFilter Group By: time(\$__interval)

Row Title	Title	Description	Query Details
		successfully executed when 1, otherwise is 0.	
Healthchecks - RDS	Entity Repository	Displays entity repository healthcheck. This healthcheck is carried out by reference data service in order to validate if it is able to reach entity repository, which is an abstraction for a non-sql db. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'entity-repository' , "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
HTTP Requests	HTTP Throughput	Measures reference-data service HTTP throughput on the selected instance.	Query A Field: "count" From: "io.dropwizard.jetty.MutableServletContextHandler.requests" Where: "host" = '\$host' , \$timeFilter Group By: time(\$interval), "host"
HTTP Requests	HTTP Throughput per Method	Measures reference-data service HTTP throughput per method on the selected instance.	Query A Field: "count" From: "io.dropwizard.jetty.MutableServletContextHandler.get-requests" Where: "host" = '\$host' , \$timeFilter Group By: time(\$interval), "host"
HTTP Requests	2XX Requests Rate	Measures reference-data service HTTP throughput, on the selected instance, with return code falls into 2xx.	Query A Field: "count" From: "io.dropwizard.jetty.MutableServletContextHandler.2xx-responses" Where: "host" = '\$host' , \$timeFilter Group By: time(\$interval), "host"
HTTP Requests	4XX Errors	Measures the number of 4xx errors returned by the selected instance.	Query A Field: "count" From: "io.dropwizard.jetty.MutableServletContextHandler.4xx-responses" Where: "host" = '\$host' , \$timeFilter Group By: time(\$interval), "host"
HTTP Requests	5XX Errors	Measures the number of 5xx errors returned by the selected instance.	Query A Field: "count" From: "io.dropwizard.jetty.MutableServletContextHandler.5xx-responses" Where: "host" = '\$host' , \$timeFilter Group By: time(\$interval), "host"fill(none)
Global Database Requests	Global Database Throughput	Measures the overall database throughput, considering all operations executed by the selected instance. It also shows a break down per operation.	Query A Field: "count" From: "reference-data-service.entity-repository" Where: "host" = '\$host' , \$timeFilter Group By: time(\$interval), "host", "operation"
Global Database Requests	Global Database Errors	Measures the total number of errors registered, on the selected instance, while performing some operations onto the database.	Query A Field: "count" From: "reference-data-service.entity-repository.error" Where: "host" = '\$host' , \$timeFilter Group By: time(\$__interval), "host"

Row Title	Title	Description	Query Details
Global Database Requests	Global Database Errors - Entities Not Found	Measures the total number of errors caused by entities not found, registered on the selected instance while performing some operations onto the database.	Query A Field: "count" From: "reference-data-service.entity-repository.entitynotfound" Where: "host" = '\$host' , \$timeFilter Group By: time(\$__interval), "host"fill(none)
Global Database Requests	Global Database Latency per Percentile for Operation	Measures global database latency at percentile 99 considering all operations executed by the selected instance.	Query A Field: "p50", "p95", "p99", "p999", "max" From: "reference-data-service.entity-repository" Where: "host" = '\$host' , operation=\$operation' , \$timeFilter Group By: time(\$__interval), "host"
Database Operation	Operation throughput for Entity	Measures the database throughput, considering the operations executed by the selected instance using the selected entity type.	Query A Field: "count" From: "reference-data-service.entity-repository" Where: "host" = '\$host' , operation=\$operation' , entityType = '\$entity' , \$timeFilter Group By: time(\$__interval), "host", "operation"
Database Operation	Operation for Entity Latency Per Percentile	Measures the latency per percentile to execute the specified operation on the selected entity type using the selected instance.	Query A Field: "p50", "p95", "p99", "p999", "max" From: "reference-data-service.entity-repository" Where: "host" = '\$host' , operation=\$operation' , entityType = '\$entity' , \$timeFilter Group By: time(\$__interval), "host", "operation"
JVM	JVM Heap	Measures the jvm heap usage of the selected instance.	Query A Field: "value" From: "letmeknow.jvm.memory.heap.used" Where: "host" = '\$host' , \$timeFilter Group By: time(\$__interval) Query B Field: "value" From: "letmeknow.jvm.memory.heap.max" Where: "host" = '\$host' , \$timeFilter Group By: time(\$__interval), "host"
JVM	JVM Heap Used (By Gen)	For the selected instance, considering the heap used, displays the usage of each generation.	Query A Field: "value" From: "letmeknow.jvm.memory.pools.G1-Old-Gen.used" Where: "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
JVM	JVM Heap Used After GC (By Gen)	For the selected instance, considering the heap used, displays the usage of each generation after gc.	Query A Field: "value" From: "letmeknow.jvm.memory.pools.G1-Old-Gen.used-after-gc" Where: "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
JVM	JVM Heap Usage After GC	Measures the jvm heap usage after gc on the selected instance.	Query A Field: "value" From: "letmeknow.jvm.memory.heap.after-gc.usage" Where: "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
JVM	JVM Garbage Collector Time	Measures the JVM GC duration on the selected instance. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	Query A Field: "value" From: "letmeknow.jvm.gc.G1-Old-Generation.time" Where: "host" = '\$host' , \$timeFilter Group By: N/A

Row Title	Title	Description	Query Details
JVM	JVM Active Threads Count	Measures the current number of active threads in the JVM on the selected instance.	Query A Field: "value" From: "letmeknow.jvm.threads.count" Where: "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
JVM	JVM File Descriptors Ratio	Measures the jvm file descriptors usage ratio of the selected instance.	Query A Field: "value" From: "letmeknow.jvm.filedescriptor" Where: "host" = '\$host' , \$timeFilter Group By: time(\$__interval)